

Mission Integrity System (MIS) — V1.2

Scenario: Compound Degradation

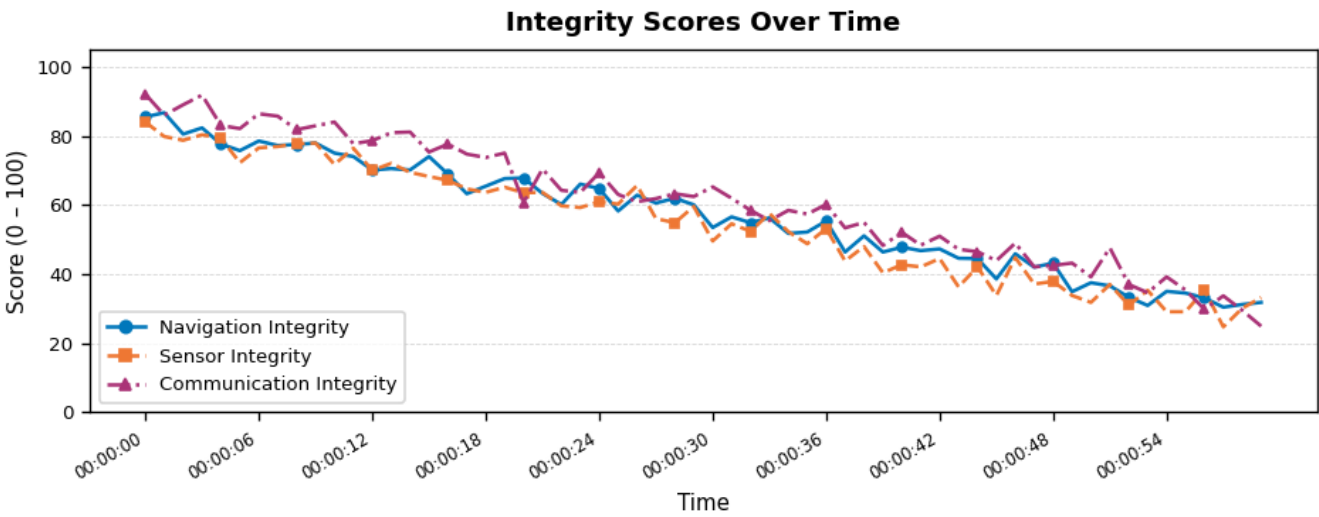
Generated: 2026-05-04 14:19:27 UTC

Demonstration report generated from simulated data

Scenario Summary

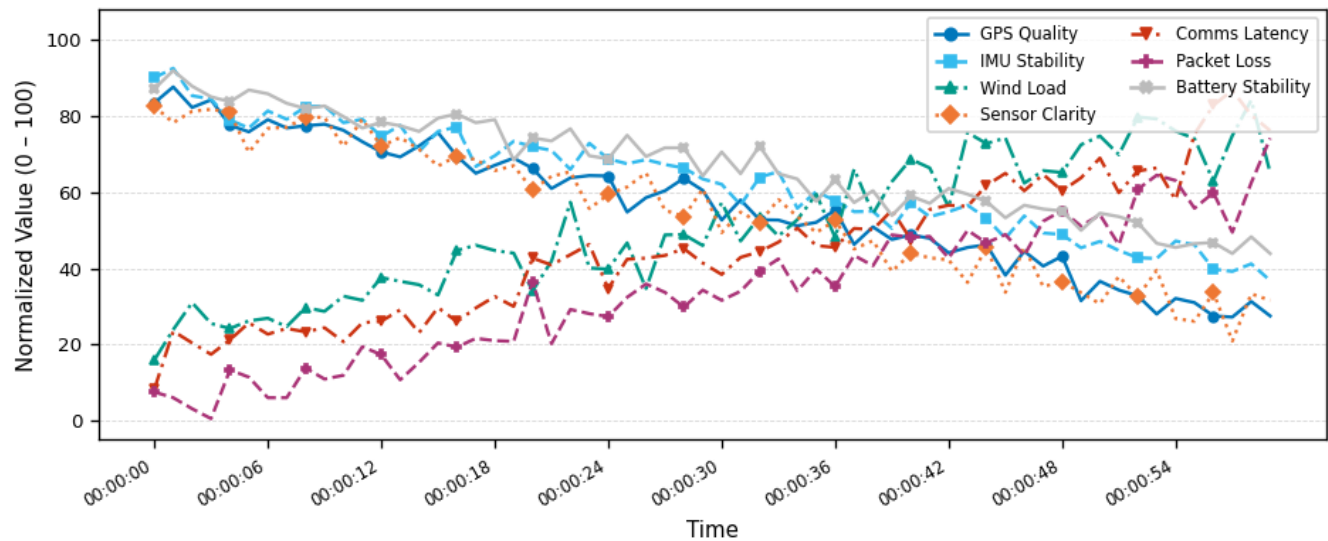
Field	Value
Final State	Unreliable
Navigation Integrity	31.8 / 100
Sensor Integrity	33.2 / 100
Communication Integrity	24.9 / 100
Causal Explanation	Unreliable: Navigation and Sensor and Communication integrity below minimum trust threshold (< 40).

Integrity Scores Over Time



Raw Signal Channels Over Time

Raw Signal Channels Over Time



Event Log

Time	Severity	Type	Message
00:00:00	INFO	State Change	Mission state initialized: Stable.
00:00:05	WARNING	State Change	State transition: Stable → Drift Emerging. Drift Emerging: Sensor integrity in drift-emerging band (60–74).
00:00:05	WARNING	Threshold Breach	Sensor integrity crossed below threshold 75: 79.4 → 72.3.
00:00:06	INFO	State Change	State transition: Drift Emerging → Stable. Stable: all integrity domains within acceptable bounds.
00:00:10	WARNING	State Change	State transition: Stable → Drift Emerging. Drift Emerging: Sensor integrity in drift-emerging band (60–74).
00:00:10	WARNING	Threshold Breach	Sensor integrity crossed below threshold 75: 78.0 → 71.8.
00:00:11	WARNING	Threshold Breach	Navigation integrity crossed below threshold 75: 75.1 → 74.1.
00:00:12	WARNING	Threshold Breach	Sensor integrity crossed below threshold 75: 76.5 → 70.1.
00:00:17	WARNING	Threshold Breach	Communication integrity crossed below threshold 75: 77.7 → 74.8.
00:00:20	WARNING	Threshold Breach	Communication integrity crossed below threshold 75: 75.1 → 60.7.
00:00:20	WARNING	Integrity Drop	Communication integrity dropped 14.4 points in one step: 75.1 → 60.7.
00:00:22	WARNING	State Change	State transition: Drift Emerging → Degraded. Degraded: Sensor integrity in degraded band (40–59).
00:00:22	WARNING	Threshold Breach	Sensor integrity crossed below threshold 60: 63.7 → 59.8.
00:00:24	WARNING	State Change	State transition: Degraded → Drift Emerging. Drift Emerging: Navigation and Sensor and Communication integrity in drift-emerging band (60–74).
00:00:25	WARNING	State Change	State transition: Drift Emerging → Degraded. Degraded: Navigation integrity in degraded band (40–59).
00:00:25	WARNING	Threshold Breach	Navigation integrity crossed below threshold 60: 64.8 → 58.3.
00:00:27	WARNING	Threshold Breach	Sensor integrity crossed below threshold 60: 65.7 → 56.1.
00:00:30	CRITICAL	State Change	State transition: Degraded → Unreliable. Unreliable: compound multi-domain degradation sustained across all domains.
00:00:30	WARNING	Threshold Breach	Navigation integrity crossed below threshold 60: 60.1 → 53.5.

Time	Severity	Type	Message
00:00:30	WARNING	Integrity Drop	Sensor integrity dropped 10.0 points in one step: 59.6 → 49.6.
00:00:32	WARNING	Threshold Breach	Communication integrity crossed below threshold 60: 62.0 → 58.5.
00:00:37	WARNING	Threshold Breach	Communication integrity crossed below threshold 60: 60.2 → 53.4.
00:00:43	CRITICAL	Threshold Breach	Sensor integrity crossed below threshold 40: 44.5 → 36.3.
00:00:45	CRITICAL	Threshold Breach	Navigation integrity crossed below threshold 40: 44.5 → 38.6.
00:00:45	CRITICAL	Threshold Breach	Sensor integrity crossed below threshold 40: 42.1 → 33.8.
00:00:47	CRITICAL	Threshold Breach	Sensor integrity crossed below threshold 40: 44.8 → 37.1.
00:00:49	CRITICAL	Threshold Breach	Navigation integrity crossed below threshold 40: 43.2 → 34.9.
00:00:50	CRITICAL	Threshold Breach	Communication integrity crossed below threshold 40: 43.2 → 39.2.
00:00:52	CRITICAL	Threshold Breach	Communication integrity crossed below threshold 40: 47.6 → 37.0.
00:00:52	CRITICAL	Integrity Drop	Communication integrity dropped 10.6 points in one step: 47.6 → 37.0.
00:00:57	CRITICAL	Integrity Drop	Sensor integrity dropped 10.7 points in one step: 35.4 → 24.7.

Data Excerpt (Last 15 Steps)

Showing the final 15 time steps of enriched scenario output. Scores are on a 0–100 scale.

Time	State	Nav	Sensor	Comms	GPS Qual	IMU Stab	Wind	Clarity	Latency	Pkt Loss	Batt Stab
00:00:45	Unreliable	38.6	33.8	43.9	38.2	47.9	74.3	33.7	65.0	48.9	53.3
00:00:46	Unreliable	45.9	44.8	49.0	44.5	53.8	62.4	45.6	60.4	43.4	56.6
00:00:47	Unreliable	42.0	37.1	42.1	40.6	49.3	65.7	35.2	64.7	52.4	55.6
00:00:48	Unreliable	43.2	37.9	42.5	43.2	48.9	65.2	36.5	60.4	55.2	55.1
00:00:49	Unreliable	34.9	33.8	43.2	31.5	45.4	72.5	33.6	63.8	51.1	50.0
00:00:50	Unreliable	37.5	31.8	39.2	36.6	47.1	74.8	30.7	69.0	54.1	54.5
00:00:51	Unreliable	36.6	37.0	47.6	34.3	44.7	69.9	37.9	59.9	46.2	53.6
00:00:52	Unreliable	33.3	31.0	37.0	32.8	42.8	79.8	32.5	65.6	60.8	52.0
00:00:53	Unreliable	30.9	35.3	34.6	28.0	42.6	79.3	39.5	66.4	64.5	46.6
00:00:54	Unreliable	35.0	29.1	39.2	32.1	47.2	76.1	26.8	58.0	63.1	45.5
00:00:55	Unreliable	34.5	29.1	35.4	31.0	46.2	74.1	26.1	75.3	55.8	46.5
00:00:56	Unreliable	33.1	35.4	29.8	27.5	39.8	62.9	33.6	82.9	59.9	46.8
00:00:57	Unreliable	30.4	24.7	33.7	27.2	39.2	74.9	20.9	86.8	49.5	43.9
00:00:58	Unreliable	31.2	30.1	29.5	31.3	41.2	84.2	33.3	80.3	62.5	48.3
00:00:59	Unreliable	31.8	33.2	24.9	27.5	36.8	65.2	31.6	76.2	74.2	43.9

Limitations and Non-Goals

Mission Integrity System (MIS) V1.2 uses simulated telemetry-like data for demonstration and portfolio purposes only. It is not a UAV control system, flight application, autopilot, navigation engine, targeting system, safety-certified tool, predictive model, or production deployment. MIS V1.2 performs deterministic classification only and does not recommend or automate operational decisions.